



Software & Documentation Overview

Overview of the Software & Documentation We're Going to Download

Software & Documentation

- CMSIS Core
- FreeRTOS
- Nucleo-64 Board User Manual
- STM32F446 Datasheet
- STM32F446 Reference Manual
- STM32 Cortex-M4 Programming Manual

CMSIS Core

CMSIS (Common Microcontroller Software Interface Standard)

- Overview: [CMSIS Documentation](#)
- CMSIS Core: [CMSIS-Core \(Cortex-M\) Documentation](#)
- Basic Project Setup: [Using CMSIS in Embedded Applications](#)
- Organization of the CMSIS Core: [CMSIS-Core Device Templates](#)

Why Use the CMSIS Core?

- Enhanced software reusability.
- Enhanced software compatibility.
- Ease of use.

FreeRTOS

General & Documentation

- Overview: [FreeRTOS Main Page](#)
- Documentation: [FreeRTOS Documentation](#)

Why Use the FreeRTOS?

- Structured Multitasking.
- Optimized Resource Management.
- Deterministic Real-Time Response.

Nucleo-64 Board User Manual

Documentation

- User Manual: [NUCLEO-64 with STM32F446RE Documents](#)

What to Use the Manual for?

- System Setup.
- Getting Started.
- Understanding Board Architecture.
- Programming & Debugging.
- Power Management.
- Indicator & Control Components.
- Clock & Communication Setup.

STM32F446 Datasheet

Documentation

- STM32F446 Datasheet: [STM32F446 - PDF Documentation](#)

What to Use the Datasheet for?

- Block Diagram Overview.
- Alternate Function Mapping.
- Clock Configuration.
- Peripheral Connection.
- Pinout & Package Information.
- Programming & Memory Organization.
- Electrical Characteristics.

STM32F446 Reference Manual

Documentation

- STM32F446 Reference Manual: [STM32F446 - PDF Documentation](#)

What to Use the Reference Manual for?

- **Peripheral Registers:**
 - Outlines how to interact directly with the hardware features of the microcontroller by manipulating control and status registers.
- **Memory Mapping:**
 - Understand the address space to properly address peripherals and memory regions during register-level programming.
- **Embedded Flash Memory & Organization:**
 - Learn how the internal flash is structured to implement efficient read/write operations and to store application code or data.

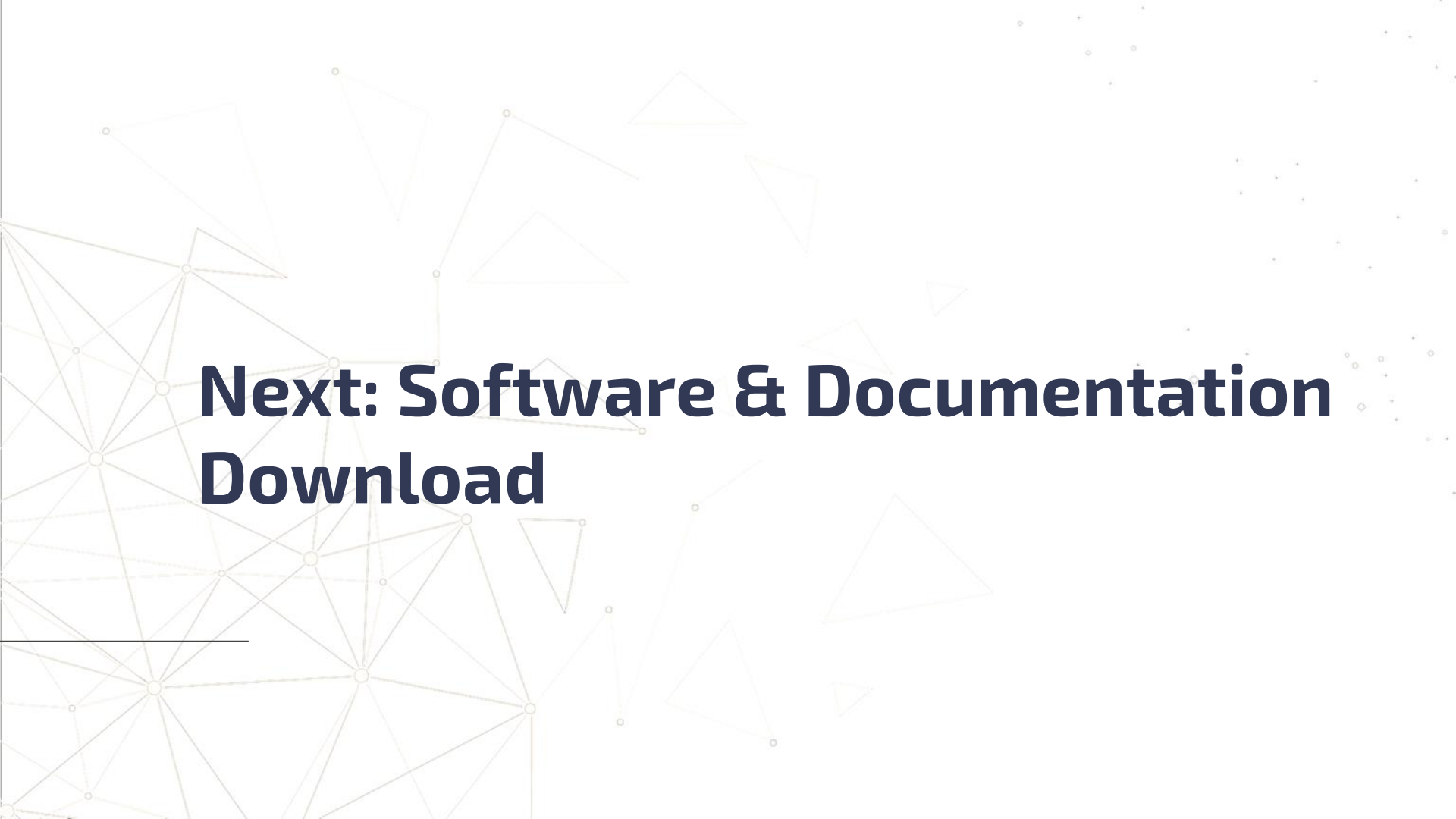
STM32 Cortex-M4 Programming Manual

Documentation

- STM32 Cortex-M4 Programming Manual: [STM32F446 - PDF Documentation](#)

What to Use the STM32 Cortex-M4 Programming Manual for?

- About the STM32 Cortex-M4 Processor & Core Peripherals.
- The Programmer's, Memory, and Exception Models, Fault Handling & Power Mgmt.
- The Instruction Set.
- Core Peripherals.

The background features a complex network of thin, light-colored lines and small circles, forming a web-like structure. Overlaid on this are several triangles of various sizes and orientations, some of which are filled with a light yellow or orange color. The overall aesthetic is modern and technical.

Next: Software & Documentation Download
